Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 2 Handout

Tuesday, September 15

Problem 2.1

Suppose every child is equally likely to be a boy or a girl, and that the gender of any newborn is independent of genders of all previously born children. A family has two children.

(a) Determine the probability that both children are boys.

(b) Determine the probability that both children are boys given that at least one child is a boy.

(c) Determine the probability that both children are boys given that least one child is a boy born on a Tuesday. Assume each child is equally likely to be born on any day of the week, independently of all other children.

Problem 2.2

Your problem set is due in 15 minutes! It is in one of n drawers, but they are messy, and you're not sure which drawer it is in. The probability that the problem set is in drawer k is p_k for $k = 1, \ldots, n$. If the problem set is in drawer k and you search there, you will find it with probability q_k for $k = 1, \ldots, n$. Suppose you searched in drawer i and did not find the problem set.

(a) Determine the probability that the problem set is in drawer j for any $j \neq i$.

(b) Determine the probability that the problem set is in drawer i.